Pet MBTI Analyzed Using Principal Component Analysis: YETI

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Abstract

This study aimed to statistically classify pet dog personalities using an MBTI-inspired framework. Based on items from the Canine Behavioral Assessment and Research Questionnaire (C-BARQ), four dichotomous axes were constructed: Extraverted/Introverted, Troublemaker/Angel, Stable/Unstable, and Bright/Clumsy, representing social openness, problem-behavior propensity, emotional stability, and learning ability, respectively. The 13,247 pets' data which were collected by Dr. James A. Serpell's team at University of Pennsylvania School of Veterinary Medicine were divided into training (70%) and validation (30%) sets. Principal component analysis (PCA) was conducted for each axis on the training set, retaining components until the cumulative explained variance exceeded 70%. Composite scores were calculated as weighted sums of standardized component scores. Outliers were excluded, and each axis was dichotomized using three candidate cutoffs. A cutoff of 0 showed the best performance across all four axes in both datasets, supporting the consistency and interpretability of the proposed classification framework.

Keywords

Principal component analysis; C-BARQ; pet personality; MBTI; YETI

1 Introduction

As pets are increasingly regarded as members of the family, both the pet industry and academic interest in pets have grown rapidly. Dogs, as one of the most representative pets, have coexisted with humans for a long time. However, canine personality and behavior cannot be fully explained by breed alone, and even within the same breed, individual differences in temperament can be substantial. In the real adoption process, adopters often select dogs based primarily on appearance or breed without sufficiently understanding each dog's temperament. When the adopter's expectations do not match the dog's actual personality, this mismatch can lead to training failure, escalation of behavioral problems, and, in many cases, relinquishment.

According to a study published in Powell et al. [11], 427 relinquished dogs from U.S. shelters were compared with 427 pet dogs from the general population. Although 69.3% of relinquishing owners reported "no behavior problems," the relinquished dogs generally showed higher problem-behavior scores on the mini C-BARQ. Significant differences were observed particularly in trainability, aggression, reactions to unfamiliar stimuli, and separation-related behaviors. These findings suggest that, rather than intentionally concealing problems, many owners may have lacked sufficient awareness or understanding of behavioral issues. Such results indicate that conducting standardized behavioral assessments in shelters and clearly providing the results to prospective adopters can be a key strategy for reducing relinquishment rates. Meanwhile, efforts to objectively understand and assess canine personality have continued across many studies. Svartberg and Forkman [14] proposed five personality dimensions in dogs—playfulness, curiosity, chase-proneness, sociability, and aggressiveness. In addition, several survey-based instruments have been developed internationally to evaluate canine personality. A representative tool

is the C-BARQ (Canine Behavioral Assessment & Research Questionnaire), which quantifies canine behavioral characteristics based on owners' questionnaire responses [3]; the instrument is available online [13]. This line of work demonstrates that academic efforts to structure and classify canine personality have accumulated over a long period.

In humans, the MBTI (Myers–Briggs Type Indicator) [1] is a widely used tool for self-understanding, career exploration, psychological counseling, and improving interpersonal relationships. By simplifying individual tendencies into 16 types, MBTI has become a practical framework that facilitates mutual understanding and communication. Jang et al. [4] reported that applying an MBTI-based counseling program to first-year medical students significantly improved interpersonal satisfaction, trust, intimacy, and openness. Another study [8] found that while MBTI did not directly predict team performance, it was effective in promoting mutual understanding and collaboration among team members. In this sense, MBTI can be viewed as a socially meaningful tool applicable to educational, counseling, and organizational settings, beyond mere personality categorization.

The MBTI framework provides a theoretical lens that can inform the structuring of canine personality in this study. In the canine domain as well, survey-based approaches to personality assessment have been actively pursued. Hsu and Serpell [3] developed a 68-item questionnaire to measure behavioral and temperament traits in pet dogs, and through factor analysis derived 11 temperament factors, including aggression, fear/anxiety, sociability, and trainability. They reported high internal consistency for many factors and significant score differences between problem-behavior groups, demonstrating that the questionnaire can be used as a reliable tool for personality/behavioral problems and to evaluate the effectiveness of behavioral interventions.

Wiener and Haskell [15] also reviewed major canine personality questionnaires, including C-BARQ and the Dog Personality Questionnaire (DPQ) [6], and suggested that owner-report survey data can serve as an efficient alternative for large-sample studies of canine personality and genetic analyses. This supports the view that standardized questionnaire tools have become a core methodology for quantifying canine personality when direct behavioral observation is difficult. In addition, the Companion Animal Mindset Indicator (CAMI) research team developed a dog personality-typing assessment by analyzing the persistence and repetitiveness of behavioral patterns, defining four dimensions (activity, relationship, adaptability, and obedience) and combining them to derive 16 personality types. This approach is meaningful in that it was designed not merely for entertainment, but to help owners understand their dogs more deeply and consider tailored parenting and training strategies. Collectively, these studies indicate sustained interest in evaluating and typing canine personality; nevertheless, an intuitive yet statistically grounded personality-categorizing method that can be directly communicated to adopters remains insufficient.

Therefore, this study proposes a novel personality type indicator that systematizes canine personality in an MBTI-like framework using C-BARQ data: the Youngseo–Eunbin Type Indicator (YETI). Specifically, we propose the following four personality axes:

- Extraverted vs. Introverted — tendency toward social interaction
- Troublemaker vs. Angel — likelihood of problem behaviors and self-regulation
- Stable vs. Unstable — emotional stability in response to unfamiliar stimuli and separation contexts

- Bright vs. Clumsy — adaptability to training and command performance

These axes are not intended as a simplistic "good vs. bad" dichotomy; rather, they present canine behavioral characteristics in a psychologically interpretable structure. Through this approach, shelters and adoption agencies can provide each dog's temperament as an intuitive and easily understandable profile, while adopters can select dogs that better fit their living environment and preferences. Ultimately, this framework may contribute to reducing post-adoption adjustment problems, preventing relinquishment, optimizing training and counseling, and fostering a culture that respects dogs as independent beings with distinct personalities.

This study is expected to make contributions across practical, social, academic, and industrial domains. On a practical level, the research aims to develop an intuitive personality-typing tool that can be integrated into shelter adoption processes, facilitating better matches between pets and adopters. From a social perspective, this tool has the potential to prevent relinquishment and promote companion-animal welfare by enhancing owner-pet compatibility. Academically, the findings may have industrial implications by enabling the expansion of personality-based customized services for companion animals, including tailored training programs, behavioral counseling, and personalized product recommendations. In other words, this study aims to move beyond evaluating dogs merely in terms of the presence or absence of problem behaviors, and to propose a new paradigm for understanding dogs as independent beings with unique personalities.

2 Materials and Methods

2.1 Data

In this study, we utilized 13,247 pets' data from the C-BARQ, a standardized survey instrument designed to assess behavioral characteristics of companion dogs. The data were collected by Dr. James A. Serpell's team at University of Pennsylvania School of Veterinary Medicine. C-BARQ is a 5-point scale in which owners rate their dogs' behaviors from 0 ("none") to 4 ("always"/"very severe"). The dataset includes both dog- and owner-related background information such as breed, sex, date of birth, body weight, neuter status, adoption source, age at evaluation, and owner demographics. The questionnaire covers a wide range of domains—including trainability, aggression, fear and anxiety, separation-related behaviors, excitability, attachment and attention-seeking, and other problem behaviors—allowing multidimensional assessment of canine behavior and temperament. These data can be used as a foundational resource to visually interpret overall behavioral profiles and to consider personality compatibility between adopters and dogs in adoption settings.

First, we reorganized the original C-BARQ items to align with four personality categories: Extraverted/Introverted, Troublemaker/Angel, Stable/Unstable, and Bright/-Clumsy. Extraverted/Introverted was defined based on a dog's level of initiative in interacting with people and environmental stimuli. Extraverted dogs were considered more affiliative and proactive toward social stimuli (e.g., humans, other dogs, and environmental changes), whereas introverted dogs were assumed to be more likely to avoid unfamiliar stimuli or exhibit anxious responses. A total of 12 items were selected for this category.

Troublemaker/Angel was designed to capture impulse control. Dogs that follow training commands well and show low distractibility were interpreted as "angel" dogs with high self-regulation, whereas dogs that frequently exhibit problem behaviors and react readily to external stimuli were categorized as "troublemaker" dogs with higher impulsivity. A total of 13 items were selected. Stable/Unstable represented emotional stability, operationalized as the level of anxiety and stress responses to changes in the external environment (e.g., unfamiliar people, separation contexts, and loud noises). Stable dogs were assumed to respond relatively calmly to unpredictable stimuli, while anxious dogs were considered more likely to display fear or anxiety. A total of 21 items were selected. Finally, Bright/Clumsy reflected learning ability and problem-solving capacity. Dogs that quickly understand and perform new tasks or commands were regarded as having high training responsiveness, whereas dogs that show slower comprehension or reduced attention were interpreted as having relatively lower learning speed. A total of 6 items were selected. The detailed list of selected items for the four personality categories is presented in Tables 1–4.

2.2 Principal Component Analysis (PCA)

This study aimed to categorize pet personality in an MBTI-like manner using multidimensional behavioral data obtained from the C-BARQ questionnaire. C-BARQ consists of dozens of items covering domains such as aggression, fear, excitability, and trainability, and thus represents a typical multivariate dataset. Beyond simply examining item-wise mean scores, we employed principal component analysis (PCA) to explore the correlation structure among items and to derive major personality factors. PCA is a widely used dimensionality-reduction technique originally proposed by Pearson [10], Hotelling [2], and it summarizes many correlated variables into a smaller set of mutually independent components.

Given p variables $X_1, X_2, \ldots, X_p$, the k-th principal component is defined as the following linear combination:

$$Z_k = a_{k1}X_1 + a_{k2}X_2 + \cdots + a_{kp}X_p, \quad k = 1, 2, \ldots, m, \tag{1}$$

where a_{kj} denotes the loading of variable X_j on the k-th principal component. The component Z_k defines a new axis that preserves as much of the variance in the original data as possible.

First, the full dataset was randomly split into a training set (70%) and a validation set (30%). Using the training set, we aggregated the items within each personality category and selected only complete cases (rows without missing values). PCA was then performed on the selected items to reduce dimensionality. For each personality category, we retained principal components until the cumulative explained variance reached approximately 70% or higher. Retaining only components that explain at least a certain proportion of variance based on cumulative explained variance is a commonly used empirical rule in PCA, and Jolliffe and Cadima [5] note that this criterion is widely adopted in practice.

2.3 Component orientation and weighting

Next, we interpreted and defined each axis based on the items with large contributions (high loadings) to the corresponding component. We then adjusted the sign of each

component to ensure consistent interpretability of personality direction. Because the sign of a principal component is mathematically arbitrary [5], we reoriented the components so that they follow intuitive directions (e.g., "higher scores indicate greater extraversion" or "higher scores indicate stronger problem-behavior tendencies").

Based on the interpreted axes, we assigned weights to the retained components using their explained-variance ratios and fixed the direction using the chosen sign vectors. For each individual, the principal component scores were standardized (z-scored) to remove scale differences, and the final personality score was computed as a weighted sum. This approach follows prior methodologies for constructing PCA-based composite indices, in which explained-variance proportions are used as component weights [e.g., 9]. Specifically, for each retained principal component, the standardized score was obtained by subtracting the training-sample mean and dividing by the corresponding training-sample standard deviation.

Let the category be indexed by $\ell \in \{\mathrm{EI}, \mathrm{TA}, \mathrm{SU}, \mathrm{BC}\}$. For category ℓ, let $x_{ik}^{(\ell)}$ denote the principal component score of individual i on the k-th component. Let K_ℓ be the set of retained components for category ℓ. Define the direction coefficient

$$s_k^{(\ell)} \in \{-1, +1\}, \quad k \in K_\ell, \tag{2}$$

and apply a sign adjustment. The weights are derived from explained variances:

$$\lambda_k^{(\ell)} = \left(\mathrm{sdev}_k^{(\ell)}\right)^2, \tag{3}$$

$$w_k^{(\ell)} = \frac{\lambda_k^{(\ell)}}{\sum_{h \in K_\ell} \lambda_h^{(\ell)}}, \quad k \in K_\ell. \tag{4}$$

Each selected principal component score is standardized within the training data:

$$z_{ik}^{(\ell)} = \frac{x_{ik}^{(\ell)} - \mu_k^{(\ell)}}{\sigma_k^{(\ell)}}, \quad k \in K_\ell, \tag{5}$$

where $\mu_k^{(\ell)}$ and $\sigma_k^{(\ell)}$ denote the mean and standard deviation, respectively, of the k-th retained principal component score in the training set. The final personality score is then computed as the weighted sum

$$S_i^{(\ell)} = \sum_{k \in K_\ell} w_k^{(\ell)} s_k^{(\ell)} z_{ik}^{(\ell)}. \tag{6}$$

2.4 Silhouette score

Individuals were classified according to their tendency using three cutoffs: 0, the mean, or the median. Among these candidate criteria, the cutoff that produced the higher silhouette score was selected and finalized as the decision rule for personality classification.

The silhouette score is a cluster validity index proposed by Rousseeuw [12], which quantifies how well each data point fits within its assigned cluster. For a given observation, the silhouette value is computed by comparing the average distance to points within the same cluster (cohesion) with the average distance to points in the nearest neighboring

cluster (separation), yielding a value between -1 and 1. Values close to 1 indicate that the observation is well matched to its own cluster and clearly separated from other clusters; values near 0 suggest that the observation lies on the boundary between clusters; and values close to -1 imply that the observation may be more appropriately assigned to a different cluster. The overall silhouette score is the mean silhouette value across all observations and is commonly used to evaluate the overall quality of clustering. In this study, we used the silhouette index to objectively assess whether the data were meaningfully partitioned, and we referred to the interpretive guidelines proposed by Kaufman and Rousseeuw [7].

$$a(i) = \frac{1}{|C(i)| - 1} \sum_{\substack{j \in C(i) \\ j \neq i}} d(i, j), \tag{7}$$

$$b(i) = \min_{C \neq C(i)} \frac{1}{|C|} \sum_{j \in C} d(i, j), \tag{8}$$

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}}, \quad -1 \leq s(i) \leq 1, \tag{9}$$

$$\bar{s} = \frac{1}{n} \sum_{i=1}^{n} s(i). \tag{10}$$

After estimating the PCA results on the training data—namely the principal component axes, standardization parameters, retained components, weights, and component directions—we applied these estimates to the validation set to compute personality scores for each individual. Specifically, the final personality score was calculated as a weighted sum of the selected principal component scores, using the weights and sign rules derived from the training set. As in the training set, we removed observations with missing values and excluded extreme values, and then computed silhouette scores to evaluate classification performance.

3 Results

The four personality dimensions considered in this study were Extraverted–Introverted (EI), Troublemaker–Angel (TA), Stable–Unstable (SU), and Bright–Clumsy (BC).

3.1 Extraverted/Introverted

PCA was conducted using the 12 items related to the Extraverted/Introverted dimension, and a total of four principal components were retained to achieve a cumulative explained variance of at least 70% (Table 5). The first component was largely driven by high loadings on items such as "is fearful when approached directly by an unfamiliar adult while away from home," "is fearful when an unfamiliar person tries to touch or pet the dog," "is fearful when unfamiliar persons visit the home," and "is fearful when approached directly by an unfamiliar child while away from home" (+0.454, +0.449, +0.433, and +0.420, respectively). Accordingly, this component was defined as an axis representing fear of unfamiliar people. Using the same procedure, the second component was interpreted

as an axis of activity and excitability, the third component as an axis of attachment toward family, and the fourth component as an axis of fear of unfamiliar situations and objects. To ensure that higher scores indicate higher extraversion, component signs were adjusted: PC3 was kept as is, whereas the signs of PC1, PC2, and PC4 were reversed. Based on these orientations and the selected components, we derived the final formula for computing the Extraversion/Introversion personality score as follows:

$$\mathrm{EI}_i = -0.449\, z_{i,\mathrm{PC1}} - 0.292\, z_{i,\mathrm{PC2}} + 0.146\, z_{i,\mathrm{PC3}} - 0.114\, z_{i,\mathrm{PC4}}. \tag{11}$$

3.2 Troublemaker/Angel

PCA was conducted using the 14 items related to the Troublemaker/Angel dimension, and a total of seven principal components were retained to achieve a cumulative explained variance of at least 70% (Table 6). The first component was mainly characterized by high loadings on items such as "will run away from home or yard if given the opportunity," "will chase or attempt to chase birds if given the opportunity," "will chase or attempt to chase small animals such as squirrels or rabbits if given the opportunity," and "will chase or attempt to chase cats if given the opportunity" (+0.367, +0.346, +0.333, and +0.329, respectively). Accordingly, this component was defined as an axis representing chasing and escape tendency. Using the same approach, the second component was interpreted as an axis of chasing impulsivity, the third component as attention/concentration, the fourth component as house-soiling accidents, the fifth component as scent-rolling tendency, and the sixth and seventh components as problem-behavior axes. To ensure that higher scores indicate a stronger "troublemaker" tendency, component signs were oriented accordingly: PC1, PC2, PC3, PC4, PC6, and PC7 were kept as estimated, while the sign of PC5 was reversed. Based on these orientations and the retained components, we derived the final formula for computing the Troublemaker/Angel personality score as follows:

$$\begin{aligned}\mathrm{TA}_i = {} & 0.328\, z_{i,\mathrm{PC1}} + 0.185\, z_{i,\mathrm{PC2}} + 0.126\, z_{i,\mathrm{PC3}} + 0.097\, z_{i,\mathrm{PC4}} \\ & - 0.095\, z_{i,\mathrm{PC5}} + 0.086\, z_{i,\mathrm{PC6}} + 0.083\, z_{i,\mathrm{PC7}}.\end{aligned} \tag{12}$$

3.3 Stable/Unstable

PCA was conducted using the 21 items related to the Stable/Unstable dimension, and a total of seven principal components were retained to achieve a cumulative explained variance of at least 70% (Table 7). The first component was primarily driven by high contributions from items such as "when approached directly by an unfamiliar adult while away from your home," "when unfamiliar persons visit your home," "when an unfamiliar person tries to touch or pet the dog," and "when an unfamiliar dog visits your home" (-0.297, -0.293, -0.292, and -0.280, respectively). Accordingly, this component was defined as an axis representing fear of unfamiliar people and dogs. Using the same procedure, the second component was interpreted as an axis of separation-related anxiety and stress, the third as vigilant/defensive responses to unfamiliar dogs approaching, the fourth as attachment anxiety, the fifth as selective attachment to specific individuals, the sixth as excessive physical reactions under anxiety, and the seventh as over-excitability and behavioral regulation. To ensure that higher scores indicate greater stability, component signs were oriented accordingly: PC1, PC3, and PC6 were retained as estimated, whereas the signs of PC2, PC4, PC5, and PC7 were reversed. Based on these orientations and

the retained components, we derived the final formula for computing the Stable/Unstable personality score as follows:

$$\begin{aligned}\mathrm{SU}_i = 0.390\, z_{i,\mathrm{PC1}} - 0.194\, z_{i,\mathrm{PC2}} + 0.113\, z_{i,\mathrm{PC3}} - 0.104\, z_{i,\mathrm{PC4}} \\ - 0.079\, z_{i,\mathrm{PC5}} + 0.065\, z_{i,\mathrm{PC6}} - 0.055\, z_{i,\mathrm{PC7}}.\end{aligned} \tag{13}$$

3.4 Bright/Clumsy

PCA was conducted using the six items related to the Bright/Clumsy dimension, and a total of three principal components were retained to achieve a cumulative explained variance of at least 70% (Table 8). The first component was largely characterized by high loadings on items such as "obeys a 'stay' command immediately," "obeys a 'sit' command immediately," "is slow to learn new tricks or tasks," and "is easily distracted by interesting sounds, smells, or visual stimuli" (+0.499, +0.475, +0.432, and +0.391, respectively). Accordingly, this component was defined as an axis of training performance and learning-related concentration. Using the same procedure, the second component was interpreted as an axis of learning and attention, and the third component as an axis of learning sensitivity. To ensure that higher scores indicate greater brightness (i.e., higher learning ability), component signs were oriented accordingly: PC1 and PC2 were retained as estimated, whereas the sign of PC3 was reversed. Based on these orientations and the retained components, we derived the final formula for computing the Bright/Clumsy personality score as follows:

$$\mathrm{BC}_i = 0.545\, z_{i,\mathrm{PC1}} + 0.253\, z_{i,\mathrm{PC2}} - 0.203\, z_{i,\mathrm{PC3}}. \tag{14}$$

3.5 Distribution and Silhouette Scores of YETI

First, the distribution of the final personality scores computed from the training data was examined using boxplots (Figure 1). Assuming that the standardized scores approximately follow a normal distribution, we excluded extreme values corresponding to roughly the upper and lower 2.5% tails by treating individuals with scores below −2 or above 2 on each personality axis as outliers and removing them from subsequent analyses.

Next, we compared two settings—without removing extreme values and after removing them—and performed dichotomous classification using three cutoffs (0, mean, or median). Interpretation of the silhouette coefficient followed the guidelines proposed by Kaufman and Rousseeuw [7]: an average silhouette value of 0.71 or higher indicates a strong cluster structure, 0.51–0.70 indicates a reasonable structure, 0.26–0.50 indicates a weak structure, and 0.25 or lower suggests no substantial cluster structure. As a result, when extreme values were removed and classification was performed using 0 as the cutoff, the silhouette scores were the highest across all four personality axes (0.533, 0.563, 0.545, and 0.535), which were all interpreted as indicating a reasonable cluster structure (Table 9). Therefore, we finalized the classification rule by removing extreme values and adopting 0 as the cutoff for the final personality classification.

Finally, because classification using 0 as the cutoff after removing outliers produced the highest silhouette scores, we adopted this rule as the final classification criterion. Using the personality score formulas derived from the training set, we computed personality scores for each category in the validation set and performed classification following the

same procedure used for the training set. Likewise, in the validation set, the distribution of the final personality scores was examined using boxplots (Figure 2).

Moreover, in the validation set as well, classification after removing outliers yielded a slightly higher silhouette score (Table 10). Because the silhouette scores were greater than 0.51 in all cases, we concluded based on the criteria suggested by Kaufman and Rousseeuw [7] that the resulting partitions show a reasonable cluster structure.

3.6 Personality Type Distribution

A total of 13,247 individuals had valid scores for all four personality dimensions, and we computed the counts and proportions for all 16 personality types. The distribution showed that EASB was the most prevalent type, accounting for 2,125 dogs (16.04%), followed by ETSB with 1,293 (9.76%), EAUB with 1,036 (7.82%), IASB with 903 (6.82%), ETUC with 894 (6.75%), ETUB with 887 (6.70%), ITUC with 797 (6.02%), ETSC with 730 (5.51%), ITUB with 661 (4.99%), ITSB with 623 (4.70%), EASC with 612 (4.62%), IAUB with 594 (4.48%), IAUC with 582 (4.39%), EAUC with 564 (4.26%), ITSC with 524 (3.96%), and IASC with 422 (3.19%).

4 Discussion

The results showed that the EASB type accounted for the largest proportion, with 2,125 dogs (16.04%). Overall, personality types characterized by extraversion, "angel" (higher self-regulation), stability, and brightness (higher learning ability) were ranked among the most prevalent, suggesting that many dogs in the sample tended to exhibit socially open, compliant, emotionally stable traits along with stronger learning-related tendencies. We additionally examined whether the proportions of personality types differed by sex, neuter status, and country (e.g., South Korea, Japan, Canada). In general, as sample sizes increased, the distributional differences became less pronounced and EASB remained the most common type. However, for countries with small sample sizes, different patterns were observed. In South Korea, although the total number of dogs was small ($n = 26$), IASB was the most frequent type (8 dogs, 30.77%). Similarly, in Japan ($n = 21$), IASC was also the most frequent (5 dogs, 23.81%). Compared with countries with larger samples, it was notable that different personality types appeared at relatively higher rates in these small-sample groups.

This study proposed a personality-typing model that avoids defining dog personality in a purely subjective manner. Instead, we transformed responses from a standardized questionnaire into single composite variables using a weighted-sum scoring approach, and then classified each dog's personality using an empirically selected decision rule. By typologizing canine personality through a statistical approach distinct from previous methods, this study offers a new framework for dog personality classification. In addition, the PCA-based scoring framework provides a reproducible statistical approach for transforming high-dimensional behavioral questionnaire data into interpretable personality indices. We expect that this framework can serve as foundational material for improving understanding of canine behavior and for enhancing the relationship between owners and their dogs.

Nevertheless, several limitations should be noted. First, the criteria used to assign questionnaire items to each personality category, as well as the interpretation of each PCA axis and the orientation of component signs, inevitably involved researcher judgment. If interpretation and classification were based more strictly on objective mathematical procedures and statistical criteria rather than subjective decisions, the resulting framework could be more reliable. Second, in the country-level comparisons, some countries had very small sample sizes, making objective inference difficult. Third, silhouette-based internal separation does not by itself establish external validity of the proposed typology.

For future work, we suggest several other directions. First, simulation studies could be conducted to evaluate whether the proposed classification rule performs robustly under a variety of conditions. Second, beyond PCA, applying factor analysis may allow items that co-vary to be grouped more objectively, reducing reliance on researcher interpretation. In addition, collecting larger and more diverse datasets across countries and breeds would enable more reliable comparisons. Longitudinal designs with repeated measurements could further examine whether an individual dog's personality type remains stable or changes over time. Furthermore, if the derived personality types can be linked to practical outcomes—such as actual behavioral problems, training methods and difficulty, owner satisfaction, or even the owner's MBTI—then YETI could evolve beyond a simple categorization into a tool with stronger real-world applicability. In this sense, the present study has value as a starting point that can be expanded and refined through follow-up research.

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CRediT authorship contribution statement

Eunbin Baek: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing–original draft, Writing–review & editing.

Youngseo Yoo: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing–original draft, Writing–review & editing.

Misung Yi: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing–review & editing.

Disclosure statement

The authors report there are no competing interests to declare.

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Data Availability Statement

The data that support the findings of this study are available from Dr. James A. Serpell at the University of Pennsylvania (serpell@vet.upenn.edu) upon reasonable request.

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Use of AI tools

An AI language model (ChatGPT, OpenAI) was used to assist with English translation and with routine calculations for reporting purposes (e.g., computing coefficients in the weighted-sum scoring formulas based on author-specified procedures). The model was not used to determine the study design, select variables, interpret results, or draw scientific conclusions. All authors reviewed and verified the outputs and take full responsibility for the final manuscript.

References

[1] Gregory J. Boyle. Myers-briggs type indicator (MBTI): some psychometric limitations. *Australian Psychologist*, 30(1):71–74, 1995. doi: 10.1111/j.1742-9544.1995.tb01750.x.

[2] Harold Hotelling. Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6):417–441, 1933. doi: 10.1037/h0071325.

[3] Yi Hsu and James A. Serpell. Development and validation of a questionnaire for measuring behavior and temperament traits in pet dogs. *Journal of the American*

Veterinary Medical Association, 223(9):1293–1300, 2003. doi: 10.2460/javma.2003.223.1293.

[4] S.S. Jang, B.J. Kim, Y.T. Ju, H.O. Woo, J.H. Han, G.J. Cho, and S.C. Hong. The effect of counseling for interpersonal relationship using myers-briggs type indicator. *Korean Journal of Medical Education*, 23(1):15–26, 2011. doi: 10.3946/kjme.2011.23.1.15.

[5] Ian T. Jolliffe and Jorge Cadima. Principal component analysis: a review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065):20150202, 2016. doi: 10.1098/rsta.2015.0202.

[6] Amanda Claire Jones. *Development and validation of a dog personality questionnaire.* PhD thesis, The University of Texas at Austin, 2008. URL `https://repositories.lib.utexas.edu/handle/2152/18189`.

[7] Leonard Kaufman and Peter J. Rousseeuw. *Finding Groups in Data: An Introduction to Cluster Analysis.* John Wiley & Sons, New York, 1990. doi: 10.1002/9780470316801.

[8] B.S. Kuipers, M.J. Higgs, N.V. Tolkacheva, and M.C. de Witte. The influence of myers-briggs type indicator profiles on team development processes: An empirical study in the manufacturing industry. *Small Group Research*, 40(4):436–464, 2009. doi: 10.1177/1046496409333938.

[9] Michela Nardo, Michaela Saisana, Andrea Saltelli, Stefano Tarantola, Anders Hoffmann, and Enrico Giovannini. *Handbook on Constructing Composite Indicators: Methodology and User Guide.* OECD Publishing, Paris, 2008. ISBN 978-92-64-04345-9. doi: 10.1787/9789264043466-en.

[10] Karl Pearson. Liii. on lines and planes of closest fit to systems of points in space. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 2(11):559–572, 1901. doi: 10.1080/14786440109462720.

[11] Lauren Powell, Deborah L. Duffy, Katherine A. Kruger, Brittany Watson, and James A. Serpell. Relinquishing owners underestimate their dog's behavioral problems: Deception or lack of knowledge? *Frontiers in Veterinary Science*, 8:734973, 2021. doi: 10.3389/fvets.2021.734973.

[12] Peter J. Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7.

[13] James A. Serpell. C-BARQ: Canine behavioral assessment & research questionnaire. Penn Veterinary Medicine, 2014. URL `https://vetapps.vet.upenn.edu/cbarq/`. Accessed: 2025-12-29.

[14] Kenth Svartberg and Björn Forkman. Personality traits in the domestic dog (Canis familiaris). *Applied Animal Behaviour Science*, 79(2):133–155, 2002. doi: 10.1016/S0168-1591(02)00121-1.

[15] Pamela Wiener and Marie J. Haskell. Use of questionnaire-based data to assess dog personality. *Journal of Veterinary Behavior*, 16:81–85, 2016. doi: 10.1016/j.jveb.2016.10.007.

Tables

Table 1: C-BARQ items included in the Extraverted/Introverted axis

C-BARQ Section	Question
Fear and Anxiety	**36**. When approached directly by an unfamiliar adult while away from your home. **37**. When approached directly by an unfamiliar child while away from your home. **39**. When unfamiliar persons visit your home. **40**. When an unfamiliar person tries to touch or pet the dog. **42**. In response to strange or unfamiliar objects on or near the sidewalk (e.g. plastic trash bags, leaves, litter, flags flapping, etc.). **47**. When first exposed to unfamiliar situations (e.g. first car trip, first time in elevator, first visit to veterinarian, etc.).
Excitability	**63**. When playing with you or other members of your household. **67**. When visitors arrive at your home.
Attachment and Attention-seeking	**70**. Tends to sit close to, or in contact with, you (or others) when you are sitting down. **71**. Tends to nudge, nuzzle or paw you (or others) for attention when you are sitting down.
Miscellaneous	**91**. Playful, puppyish, boisterous. **92**. Active, energetic, always on the go.

Table 2: C-BARQ items included in the Troublemaker/Angel axis

C-BARQ Section	Question
Training difficulty	**1**. When off the leash, returns immediately when called. **4**. Seems to attend/listen closely to everything you say or do.
Separation-related behavior	**60**. Chewing/scratching at doors, floor, windows, curtains, etc.
Miscellaneous	**74**. Chases or would chase cats given the opportunity. **75**. Chases or would chase birds given the opportunity. **76**. Chases or would chase squirrels, rabbits and other small animals given the opportunity. **77**. Escapes or would escape from home or yard given the chance. **78**. Rolls in animal droppings or other "smelly" substances. **80**. Chews inappropriate objects. **81**. "Mounts" objects, furniture, or people. **83**. Steals food. **86**. Urinates against objects/furnishings in your home. **87**. Urinates when approached, petted, handled or picked up. **97**. Barks persistently when alarmed or excited.

Table 3: C-BARQ items included in the Stable/Unstable axis

C-BARQ Section	Question
Fear and Anxiety	**36**. When approached directly by an unfamiliar adult while away from your home. **37**. When approached directly by an unfamiliar child while away from your home. **39**. When unfamiliar persons visit your home. **40**. When an unfamiliar person tries to touch or pet the dog. **45**. When approached directly by an unfamiliar dog of the same or larger size. **46**. When approached directly by an unfamiliar dog of a smaller size. **47**. When first exposed to unfamiliar situations (e.g. first car trip, first time in elevator, first visit to veterinarian, etc.). **52**. When unfamiliar dogs visit your home. **53**. When barked, growled, or lunged at by an unfamiliar dog.
Separation-related behavior	**54**. Shaking, shivering or trembling. **55**. Excessive salivation. **56**. Restlessness/agitation/pacing. **57**. Whining. **58**. Barking. **59**. Howling.
Attachment and Attention-seeking	**68**. Displays a strong attachment for one particular member of the household. **69**. Tends to follow you (or other members of household) about the house, from room to room. **72**. Becomes agitated (whines, jumps up, tries to intervene) when you (or others) show affection for another person. **73**. Becomes agitated (whines, jumps up, tries to intervene) when you show affection for another dog or animal.
Miscellaneous	**90**. Hyperactive, restless, has trouble settling down. **97**. Barks persistently when alarmed or excited.

Table 4: C-BARQ items included in the Bright/Clumsy axis

C-BARQ Section	Question
Training difficulty	**2**. Obeys the "sit" command immediately. **3**. Obeys the "stay" command immediately. **5**. Slow to respond to correction or punishment; "thick-skinned". **6**. Slow to learn new tricks or tasks. **7**. Easily distracted by interesting sights, sounds or smells.
Miscellaneous	**93**. Stares intently at nothing visible.

Table 5: PCA results for the Extraverted/Introverted axis

	PC1	PC2	PC3	PC4
Standard deviation	1.946	1.570	1.110	0.980
Proportion of Variance	0.316	0.205	0.103	0.080
Cumulative Proportion	0.316	0.521	0.624	0.704

Table 6: PCA results for the Troublemaker/Angel axis

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.813	1.359	1.124	0.984	0.975	0.929	0.916
Proportion of Variance	0.235	0.132	0.090	0.069	0.068	0.062	0.060
Cumulative Proportion	0.235	0.367	0.457	0.526	0.594	0.656	0.716

Table 7: PCA results for the Stable/Unstable axis

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.410	1.701	1.307	1.248	1.085	0.990	0.907
Proportion of Variance	0.277	0.138	0.081	0.074	0.056	0.047	0.039
Cumulative Proportion	0.277	0.414	0.496	0.570	0.626	0.673	0.712

Table 8: PCA results for the Bright/Clumsy axis

	PC1	PC2	PC3
Standard deviation	1.505	1.021	0.919
Proportion of Variance	0.378	0.174	0.141
Cumulative Proportion	0.378	0.552	0.692

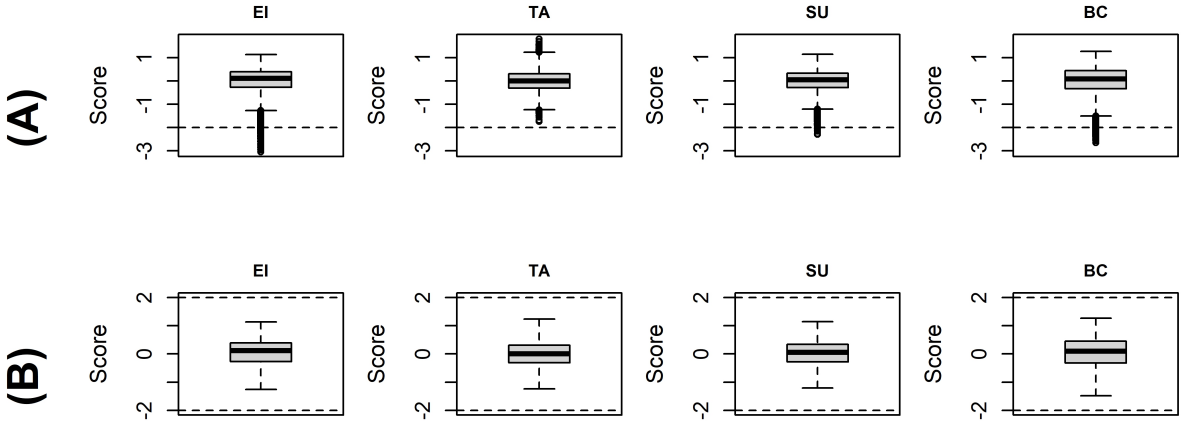
Table 9: Silhouette scores under different extreme-value handling and cutoffs (training set)

Dataset	Setting	EI	TA	SU	BC
Training	Before removal, cutoff 0	0.526	0.563	0.545	0.527
	Before removal, cutoff mean	0.526	0.563	0.545	0.527
	Before removal, cutoff median	0.505	0.563	0.539	0.515
	After removal, cutoff 0	**0.533**	**0.563**	**0.545**	**0.535**
	After removal, cutoff mean	0.531	0.563	0.545	0.533
	After removal, cutoff median	0.511	0.563	0.540	0.522

Table 10: Silhouette scores under different extreme-value handling and cutoffs (validation set)

Dataset	Setting	EI	TA	SU	BC
Validation	Before removal, cutoff 0	0.533	0.561	0.544	0.529
	After removal, cutoff 0	0.533	0.561	0.545	0.538

Figures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(B)
EI
TA
SU
BC
Score
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-1
1
-2
0
2

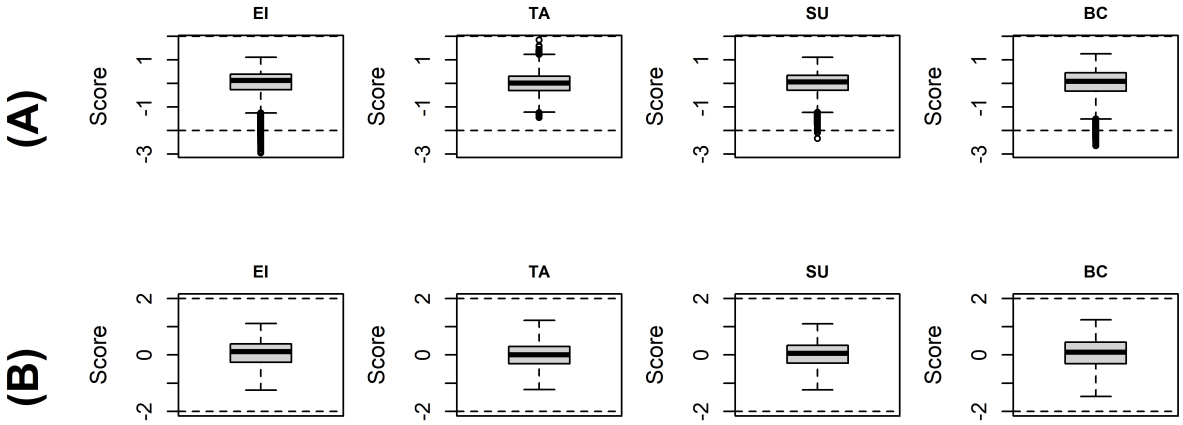
(A)
(B)
EI
TA
SU
BC
Score
-3
-1
1
-2
0
2

Figure captions

Figure 1. Boxplots of personality scores in the training set before (A) and after (B) removing extreme values.

Figure 2. Boxplots of personality scores in the validation set before (A) and after (B) removing extreme values.